

B. Tech Degree VI Semester Examination April 2011

ME 606 CAD/CAM (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain Computer Aided Design
(b) Discuss various types of transfer mechanisms used to move parts between stations.
(c) Explain any five advantages of Numerical Control.
(d) What are the miscellaneous functions? Explain any two of them.
(e) Write a note on Computer Numerical Control Systems.
(f) Explain Automatic Tool changing system.
(g) Explain any five applications of robots.
(h) Explain the concept of Flexible Manufacturing System.

PART – B

(4 x 15 = 60)

- II. (a) What is automation? What are the levels of automation? (10)
(b) Explain Detroit type of automation. (5)
- OR**
- III. Discuss the techniques for solving large scale line balancing problems based on the use of the computer with examples. Also discuss the other methods to improve line balance. (15)
- IV. (a) Differentiate between straight and contouring systems. (7)
(b) Differentiate between open loop and closed loop systems. (8)
- OR**
- V. (a) Discuss about the computer assisted part programming. (8)
(b) Write notes on geometry statements in APT. (7)
- VI. (a) Explain basic forms of CNC controlled loading system. (12)
(b) What are the advantages of CNC-Controlled loading system? (3)
- OR**
- VII. Explain special tools and tool systems in CNC Systems for lathes. Also state the advantages of CNC controlled production system. (15)
- VIII. (a) Explain the control approaches of robots. (8)
(b) Discuss the classification of robots. (7)
- OR**
- IX. Write short notes on :
(i) Sensors in robots
(ii) Artificial intelligence
(iii) Accuracy and Repeatability. (15)